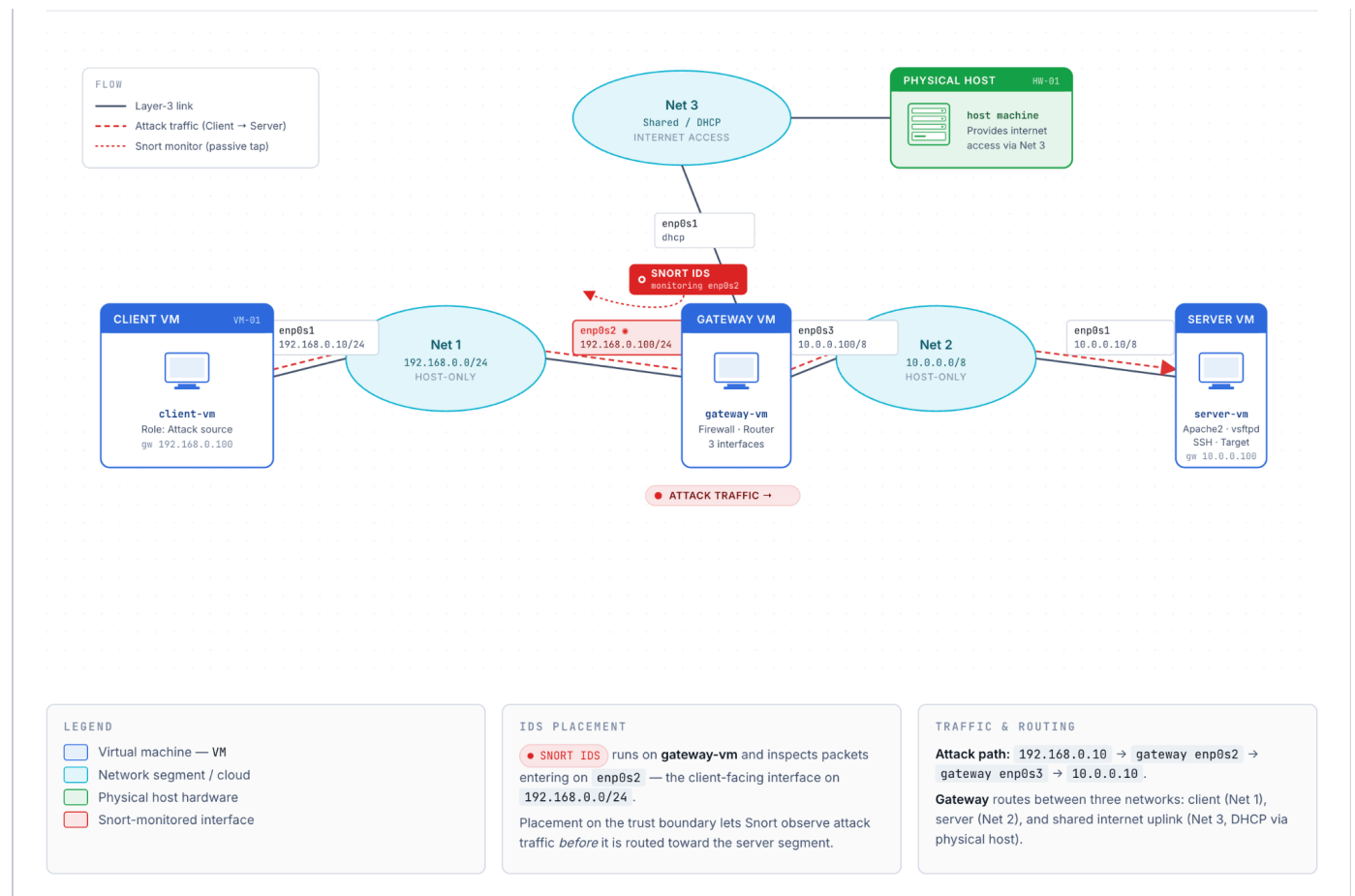


I. PROJECT OVERVIEW

This project involved setting up a Network Intrusion Detection System (NIDS) using Snort on a Gateway VM to detect and log malicious network attacks. The project required configuring Snort with custom detection rules, deploying five different types of network attacks (Land Attack, SYN Flood, Smurf Attack, UDP Flood, and five Port Scanning techniques), and verifying that Snort successfully detected each attack. The Gateway VM was configured to monitor traffic on the client-facing interface (enp0s2) with all firewall rules disabled to allow attack traffic to pass through for detection. Nine custom Snort rules were created in `/etc/snort/rules/local.rules` to identify each specific attack pattern. All attacks were successfully deployed from the Client VM using `hping3` and `nmap`, and Snort generated appropriate alerts demonstrating the effectiveness of signature-based intrusion detection.

II. NETWORK SETUP



Network Configurations

VM	Interface	IP Address	Network / Role
Client VM	enp0s1	192.168.0.10/24	Net1 - Internal client network; default gateway: 192.168.0.100

Gateway VM	enp0s2	192.168.0.100/24	Net1 - Client-facing interface (Snort monitoring)
Gateway VM	enp0s3	10.0.0.100/8	Net2 - Server-facing interface
Gateway VM	enp0s1	DHCP assigned	Net3 - Internet access via Shared Network
Server VM	enp0s1	10.0.0.10/8	Net2 - Internal server network; default gateway: 10.0.0.100

DNS Servers

All VMs were configured with the following DNS nameservers: 8.8.8.8 (Google Primary) and 8.8.4.4 (Google Secondary).

Initial Reachability

Before implementing Snort detection, all VMs could reach each other through the Gateway. The Gateway was configured with IP forwarding enabled and firewall rules set to ACCEPT (blacklist mode) to allow all attack traffic to pass through for Snort detection and logging.

III. SOFTWARE

Software	Purpose
Snort	Network Intrusion Detection System (NIDS) running on Gateway VM to detect and alert on malicious traffic patterns using signature-based detection rules
hping3	Network testing tool used on Client VM to craft and send custom TCP, UDP, and ICMP packets for deploying DoS attacks and port scans
nmap	Network scanning tool used on Client VM for TCP Null port scanning
Apache2	Web server running on Server VM, used as target for HTTP-based attacks
vsftpd	FTP server running on Server VM, configured for anonymous access
OpenSSH	SSH server running on Server VM for remote access

IV. PROJECT DESCRIPTION

Note: All terminal screenshots display the hostname 'anujtr@client-vm' regardless of which VM the command was executed on, as the Server VM and Gateway VM were created by cloning the Client VM and the hostname was not changed.

Task 1: Set Up Lab Environment

Task 1.1: Gateway Configuration

[1] Disabled firewall and set iptables to ACCEPT all traffic (blacklist mode) on Gateway VM:

```
$ sudo ufw disable
$ sudo iptables -F
$ sudo iptables -X
$ sudo iptables -t nat -F
$ sudo iptables -t nat -X
$ sudo iptables -P INPUT ACCEPT
$ sudo iptables -P OUTPUT ACCEPT
$ sudo iptables -P FORWARD ACCEPT
```

[2] Enabled IP forwarding on Gateway VM:

```
$ sudo sysctl -w net.ipv4.ip_forward=1
```

[3] Disabled reverse path filtering to allow spoofed packets (required for Land and Smurf attacks):

```
$ sudo sysctl -w net.ipv4.conf.default.rp_filter=0
$ sudo sysctl -w net.ipv4.conf.all.rp_filter=0
$ sudo sysctl -w net.ipv4.conf.all.accept_redirects=1
$ sudo sysctl -w net.ipv4.conf.all.send_redirects=1
```

[4] Enabled broadcast ping response on Gateway (required for Smurf attack):

```
$ sudo sysctl -w net.ipv4.icmp_echo_ignore_broadcasts=0
```

[5] Made configuration changes persistent by adding to /etc/sysctl.conf:

```
net.ipv4.ip_forward = 1
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.all.rp_filter = 0
net.ipv4.conf.all.accept_redirects = 1
net.ipv4.conf.all.send_redirects = 1
```

```
net.ipv4.icmp_echo_ignore_broadcasts = 0
```

```
anujtr@client-vm:~$ sudo iptables -L | grep policy
Chain INPUT (policy ACCEPT)
Chain FORWARD (policy ACCEPT)
Chain OUTPUT (policy ACCEPT)
anujtr@client-vm:~$ _
```

Task 1.2: Install Snort on Gateway

[6] Installed Snort on Gateway VM:

```
$ sudo apt-get update -y
$ sudo apt-get install -y snort
```

[7] During installation, configured Snort with:

- Address range for local network: 10.0.0.0/8 (server-side HOME_NET)
- Interface to monitor: enp0s2 (client-facing interface)

[8] Verified Snort installation:

```
$ snort --version
```

```
anujtr@client-vm:~$ snort --version
```

```

  ____  _
 / ___|| | | |
| |___| |_| |
 \___|_____|_|_|

-*> Snort! <*-
Version 2.9.15.1 GRE (Build 15125)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
```

S

Task 1.3: Configure Snort

[9] Backed up original Snort configuration:

```
$ sudo cp /etc/snort/snort.conf /etc/snort/snort.conf.backup
```

[10] Edited /etc/snort/snort.conf to set network variables:

```
ipvar HOME_NET 10.0.0.0/8
ipvar EXTERNAL_NET !$HOME_NET
```

[11] Commented out all default rule includes to use only custom rules:

```
$ sudo sed -i 's/^include \$RULE_PATH/#include \$RULE_PATH/' /etc/snort/snort.conf
```

[12] Enabled local.rules for custom detection rules:

```
include $RULE_PATH/local.rules
```

[13] Created empty local.rules file:

```
$ sudo touch /etc/snort/rules/local.rules
```

[14] Tested Snort configuration:

```
$ sudo snort -T -c /etc/snort/snort.conf
```

```
Snort successfully validated the configuration!
Snort exiting
```

Task 1.4: Install Attack Tools on Client

[15] Enabled temporary internet access for Client VM by configuring NAT on Gateway:

```
$ sudo iptables -t nat -A POSTROUTING -s 192.168.0.0/24 -o enp0s1 -j MASQUERADE
```

```
$ sudo iptables -A FORWARD -i enp0s2 -o enp0s1 -j ACCEPT
```

```
$ sudo iptables -A FORWARD -i enp0s1 -o enp0s2 -m state --state ESTABLISHED,RELATED -j ACCEPT
```

[16] Installed hping3 and nmap on Client VM:

```
$ sudo apt-get update -y
```

```
$ sudo apt-get install -y hping3 nmap
```

[17] Verified installations:

```
$ hping3 --version
```

```
$ nmap --version
```

```
anujtr@client-vm:~$ hping3 --version && nmap --version
hping3 version 3.0.0-alpha-2 ($Id: release.h,v 1.4 2004/04/09 23:38:56 antirez Exp $)
This binary is TCL scripting capable
Nmap version 7.80 ( https://nmap.org )
```

Task 2: Create Snort Detection Rules

[18] Created nine custom Snort rules in /etc/snort/rules/local.rules to detect each attack type. All rules use SID (Snort ID) greater than 10,000,000 as required for local custom rules.

Complete local.rules file:

```
# Task 4.1: Land Attack Detection
```

```
alert tcp 10.0.0.10 80 -> 10.0.0.10 80 (msg:"LAND ATTACK DETECTED - Same Source and Destination"; flags:S; sid:10000001; rev:1;)
```

```
# Task 4.2: SYN Flood Attack Detection
```

```
alert tcp any any -> $HOME_NET any (msg:"SYN FLOOD ATTACK DETECTED"; flags:S; threshold: type both, track by_dst, count 20, seconds 60; sid:10000002; rev:1;)
```

```
# Task 4.3: Smurf Attack Detection
```

```
alert icmp any any -> 192.168.0.255 any (msg:"SMURF ATTACK DETECTED - ICMP to Broadcast"; itype:8; sid:10000003; rev:1;)
```

```
# Task 4.4: UDP Flood Attack Detection
```

```
alert udp any any -> $HOME_NET any (msg:"UDP FLOOD ATTACK DETECTED"; threshold: type both, track by_dst, count 10, seconds 60; sid:10000004; rev:1;)
```

```
# Task 4.5a: TCP ACK Scan Detection
```

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP ACK SCAN DETECTED"; flags:A; threshold: type both, track by_src, count 20, seconds 60; sid:10000005; rev:1;)
```

```
# Task 4.5b: TCP FIN Scan Detection
```

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP FIN SCAN DETECTED"; flags:F; threshold: type both, track by_src, count 20, seconds 60; sid:10000006; rev:1;)
```

```
# Task 4.5c: TCP Xmas Scan Detection
```

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP XMAS SCAN DETECTED"; flags:FPU;
threshold: type both, track by_src, count 20, seconds 60; sid:10000007; rev:1;)
```

```
# Task 4.5d: TCP Null Scan Detection
```

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP NULL SCAN DETECTED"; flags:0;
threshold: type both, track by_src, count 20, seconds 60; sid:10000008; rev:1;)
```

```
# Task 4.5e: UDP Port Scan Detection
```

```
alert udp any any -> $HOME_NET any (msg:"PORT SCAN - UDP SCAN DETECTED"; threshold: type
both, track by_src, count 20, seconds 60; sid:10000009; rev:1;)
```

```
anujtr@client-vm:~$ cat /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.

#Task 4.1: Land Attack Detection
alert tcp 10.0.0.10 80 -> 10.0.0.10 80 (msg:"LAND ATTACK DETECTED - Same Source and Destination"; flags:S; sid:10000001; rev:1;)

#Task 4.2: SYN Flood Attack Detection
alert tcp any any -> $HOME_NET any (msg:"SYN FLOOD ATTACK DETECTED"; flags:S; threshold: type both, track by_dst, count 20, seconds 60; sid:10000002; rev:1;)

#Task 4.3: Smurf Attack Detection
alert icmp any any -> 192.168.0.255 any (msg:"SMURF ATTACK DETECTED - ICMP to Broadcast"; itype:8; sid:10000003; rev:1;)

#Task 4.4: UDP Flood Attack Detection
alert udp any any -> $HOME_NET any (msg:"UDP FLOOD ATTACK DETECTED"; threshold: type both, track by_dst, count 10, seconds 60; sid:10000004; rev:1;)

#Task 4.5a: TCP ACK Scan Detection
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP ACK SCAN DETECTED"; flags:A; threshold: type both, track by_src, count 20, seconds 60; sid:10000005; rev:1;)

#Task 4.5b: TCP FIN Scan Detection
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP FIN SCAN DETECTED"; flags:F; threshold: type both, track by_src, count 20, seconds 60; sid:10000006; rev:1;)

#Task 4.5c: TCP Xmas Scan Detection
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP XMAS SCAN DETECTED"; flags:FPU; threshold: type both, track by_src, count 20, seconds 60; sid:10000007; rev:1;)

#Task 4.5d: TCP Null Scan Detection
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP NULL SCAN DETECTED"; flags:0; threshold: type both, track by_src, count 20, seconds 60; sid:10000008; rev:1;)

#Task 4.5e: UDP Port Scan Detection
alert udp any any -> $HOME_NET any (msg:"PORT SCAN - UDP SCAN DETECTED"; threshold: type both, track by_src, count 20, seconds 60; sid:10000009; rev:1;)
```

[19] Validated Snort configuration with new rules:

```
$ sudo snort -T -c /etc/snort/snort.conf
```

Task 4.1: Land Attack Detection

Attack Description

A Land attack sends spoofed TCP SYN packets where the source IP address and port are identical to the destination IP address and port. This creates a connection where the host appears to be connecting to itself, potentially causing the system to crash or hang due to the confusion in handling the self-referencing connection. The attack exploits poor TCP/IP stack implementations that do not properly validate source and destination addresses.

Snort Detection Rule

```
alert tcp 10.0.0.10 80 -> 10.0.0.10 80 (msg:"LAND ATTACK DETECTED - Same Source and Destination"; flags:S; sid:10000001; rev:1;)
```

Rule Explanation:

- alert tcp - Generate alert for TCP packets
- 10.0.0.10 80 -> 10.0.0.10 80 - Source and destination are identical (IP:port)
- flags:S - TCP SYN flag must be set
- sid:10000001 - Unique Snort rule ID

Attack Execution

[20] Started Snort in detection mode on Gateway VM:

```
$ sudo snort -A console -c /etc/snort/snort.conf -i enp0s2 -q
```

[21] Deployed Land attack from Client VM:

```
$ sudo hping3 -S 10.0.0.10 -a 10.0.0.10 -p 80 -s 80 -c 5
```

Command Explanation:

- -S: Set TCP SYN flag
- -a 10.0.0.10: Spoof source IP as server IP
- -p 80: Destination port 80
- -s 80: Source port 80
- -c 5: Send 5 packets

```
anujtr@client-vm:~$ sudo hping3 -S 10.0.0.10 -a 10.0.0.10 -p 80 -s 80 -c 5
HPING 10.0.0.10 (enp0s1 10.0.0.10): S set, 40 headers + 0 data bytes
len=40 ip=10.0.0.10 ttl=64 id=60669 sport=80 flags=S seq=0 win=512 rtt=7.7 ms

--- 10.0.0.10 hping statistic ---
5 packets transmitted, 1 packets received, 80% packet loss
round-trip min/avg/max = 7.7/7.7/7.7 ms
```

Detection Results

[22] Snort successfully detected the Land attack and generated alerts showing the same source and destination IP:port combination.

```
anujtr@client-vm:~$ sudo snort -A console -c /etc/snort/snort.conf -l /var/log/snort -h 10.0.0.0/8 -q -i enp0s2
05/01-21:06:16.323702  [**] [1:10000001:1] LAND ATTACK DETECTED - Same Source and Destination [**] [Priority: 0] {TCP} 10.0.0.10:80 -> 10.0.0.10:80
```

Result: Land attack successfully detected. Snort identified TCP packets with identical source and destination (10.0.0.10:80 -> 10.0.0.10:80) and SYN flag set.

Task 4.2: SYN Flood Attack Detection

Attack Description

A SYN Flood attack exploits the TCP three-way handshake by sending a large volume of TCP SYN packets, often with spoofed source IP addresses. The target server responds with SYN-ACK packets and allocates resources to maintain half-open connections, but the final ACK never arrives. This exhausts the server's connection queue, preventing legitimate users from establishing connections.

Snort Detection Rule

```
alert tcp any any -> $HOME_NET any (msg:"SYN FLOOD ATTACK DETECTED"; flags:S; threshold:
type both, track by_dst, count 20, seconds 60; sid:10000002; rev:1;)
```

Attack Execution

[23] Deployed SYN Flood attack from Client VM:

```
$ sudo hping3 -a 192.168.0.50 10.0.0.10 -S -p 80 --faster -c 50
```

```
anujtr@client-vm:~$ sudo hping3 -a 192.168.0.50 10.0.0.10 -S -p 80 --faster -c 50
HPING 10.0.0.10 (enp0s1 10.0.0.10): S set, 40 headers + 0 data bytes
```

```
--- 10.0.0.10 hping statistic ---
```

```
50 packets transmitted, 0 packets received, 100% packet loss
```

```
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
05/01-21:07:50.766798  [**] [1:10000002:1] SYN FLOOD ATTACK DETECTED [**] [Priority: 0] {TCP} 192.168.0.50:1739 -> 10.0.0.10:80
```

Result: SYN Flood attack successfully detected.

Task 4.3: Smurf Attack Detection

Attack Description

A Smurf attack is a DDoS attack that exploits ICMP echo request and broadcast addressing. The attacker sends ICMP packets to a broadcast address with the victim's spoofed source IP, creating an amplification effect.

Snort Detection Rule

```
alert icmp any any -> 192.168.0.255 any (msg:"SMURF ATTACK DETECTED - ICMP to Broadcast"; itype:8; sid:10000003; rev:1;)
```

Attack Execution

[24] Deployed Smurf attack from Client VM:

```
$ sudo hping3 -1 --flood -a 10.0.0.10 192.168.0.255 -c 5
```

```
anujtr@client-vm:~$ sudo hping3 -1 --flood -a 10.0.0.10 192.168.0.255 -c 5
HPING 192.168.0.255 (enp0s1 192.168.0.255): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

```
05/01-21:09:21.995595  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995595  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995598  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995601  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995601  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
05/01-21:09:21.995601  [**] [1:10000003:1] SMURF ATTACK DETECTED - ICMP to Broadcast [**] [Priority: 0] {ICMP} 10.0.0.10 -> 192.168.0.255
```

Result: Smurf attack successfully detected with amplification effect visible.

Task 4.4: UDP Flood Attack Detection

Attack Description

A UDP Flood attack sends a large volume of UDP packets to the target system, overwhelming network capacity and processing resources.

Snort Detection Rule

```
alert udp any any -> $HOME_NET any (msg:"UDP FLOOD ATTACK DETECTED"; threshold: type
both, track by_dst, count 10, seconds 60; sid:100000004; rev:1;)
```

Attack Execution

[25] Deployed UDP Flood attack from Client VM:

```
$ sudo hping3 -2 --flood -a 192.168.0.50 10.0.0.10 -c 50
```

```
anujtr@client-vm:~$ sudo hping3 -2 --flood -a 192.168.0.50 10.0.0.10 -c 5
HPING 10.0.0.10 (enp0s1 10.0.0.10): udp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

```
05/01-21:11:00.709123  [**] [1:100000004:1] UDP FLOOD ATTACK DETECTED [**] [Priority: 0] {UDP} 192.168.0.50:2453 -> 10.0.0.10:0
05/01-21:11:00.709155  [**] [1:100000009:1] PORT SCAN - UDP SCAN DETECTED [**] [Priority: 0] {UDP} 192.168.0.50:2463 -> 10.0.0.10:0
```

Result: UDP Flood attack successfully detected.

Task 4.5: Port Scanning Detection

Task 4.5a: TCP ACK Scan

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP ACK SCAN DETECTED"; flags:A;
threshold: type both, track by_src, count 20, seconds 60; sid:10000005; rev:1;)
```

[26] Attack command:

```
$ sudo hping3 -A 10.0.0.10 -p ++1 -c 30
```

```
anujtr@client-vm:~$ sudo hping3 -A 10.0.0.10 -p ++1 -c 30
HPING 10.0.0.10 (enp0s1 10.0.0.10): A set, 40 headers + 0 data bytes
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=1 flags=R seq=0 win=0 rtt=3.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=2 flags=R seq=1 win=0 rtt=2.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=3 flags=R seq=2 win=0 rtt=7.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=4 flags=R seq=3 win=0 rtt=7.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=5 flags=R seq=4 win=0 rtt=6.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=6 flags=R seq=5 win=0 rtt=5.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=7 flags=R seq=6 win=0 rtt=2.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=8 flags=R seq=7 win=0 rtt=5.7 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=9 flags=R seq=8 win=0 rtt=2.3 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=10 flags=R seq=9 win=0 rtt=8.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=11 flags=R seq=10 win=0 rtt=6.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=12 flags=R seq=11 win=0 rtt=4.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=13 flags=R seq=12 win=0 rtt=5.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=14 flags=R seq=13 win=0 rtt=2.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=15 flags=R seq=14 win=0 rtt=6.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=16 flags=R seq=15 win=0 rtt=2.7 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=17 flags=R seq=16 win=0 rtt=7.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=18 flags=R seq=17 win=0 rtt=4.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=19 flags=R seq=18 win=0 rtt=1.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=20 flags=R seq=19 win=0 rtt=9.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=21 flags=R seq=20 win=0 rtt=6.3 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=22 flags=R seq=21 win=0 rtt=5.7 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=23 flags=R seq=22 win=0 rtt=4.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=24 flags=R seq=23 win=0 rtt=5.2 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=25 flags=R seq=24 win=0 rtt=7.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=26 flags=R seq=25 win=0 rtt=3.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=27 flags=R seq=26 win=0 rtt=2.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=28 flags=R seq=27 win=0 rtt=5.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=29 flags=R seq=28 win=0 rtt=1.7 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=30 flags=R seq=29 win=0 rtt=6.8 ms

--- 10.0.0.10 hping statistic ---
30 packets transmitted, 30 packets received, 0% packet loss
round-trip min/avg/max = 1.6/5.0/9.6 ms
```

```
05/01-21:12:18.271358  [**] [1:10000005:1] PORT SCAN - TCP ACK SCAN DETECTED [**] [Priority: 0] {TCP} 192.168.0.10:1847 -> 10.0.0.10:20
```

Task 4.5b: TCP FIN Scan

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP FIN SCAN DETECTED"; flags:F;
threshold: type both, track by_src, count 20, seconds 60; sid:10000006; rev:1;)
```

[27] Attack command:

```
$ sudo hping3 -F 10.0.0.10 -p ++1 -c 30
```

```
anujtr@client-vm:~$ sudo hping3 -F 10.0.0.10 -p ++1 -c 30
HPING 10.0.0.10 (enp0s1 10.0.0.10): F set, 40 headers + 0 data bytes
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=1 flags=RA seq=0 win=0 rtt=3.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=2 flags=RA seq=1 win=0 rtt=9.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=3 flags=RA seq=2 win=0 rtt=6.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=4 flags=RA seq=3 win=0 rtt=4.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=5 flags=RA seq=4 win=0 rtt=10.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=6 flags=RA seq=5 win=0 rtt=7.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=7 flags=RA seq=6 win=0 rtt=5.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=8 flags=RA seq=7 win=0 rtt=10.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=9 flags=RA seq=8 win=0 rtt=6.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=10 flags=RA seq=9 win=0 rtt=4.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=11 flags=RA seq=10 win=0 rtt=8.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=12 flags=RA seq=11 win=0 rtt=9.3 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=13 flags=RA seq=12 win=0 rtt=8.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=14 flags=RA seq=13 win=0 rtt=7.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=15 flags=RA seq=14 win=0 rtt=7.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=16 flags=RA seq=15 win=0 rtt=5.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=17 flags=RA seq=16 win=0 rtt=4.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=18 flags=RA seq=17 win=0 rtt=10.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=19 flags=RA seq=18 win=0 rtt=6.2 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=20 flags=RA seq=19 win=0 rtt=1.7 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=23 flags=RA seq=22 win=0 rtt=5.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=24 flags=RA seq=23 win=0 rtt=5.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=25 flags=RA seq=24 win=0 rtt=2.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=26 flags=RA seq=25 win=0 rtt=1.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=27 flags=RA seq=26 win=0 rtt=8.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=28 flags=RA seq=27 win=0 rtt=8.3 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=29 flags=RA seq=28 win=0 rtt=4.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=30 flags=RA seq=29 win=0 rtt=9.2 ms

--- 10.0.0.10 hping statistic ---
30 packets transmitted, 28 packets received, 7% packet loss
round-trip min/avg/max = 1.7/6.6/10.6 ms
```

```
05/01-21:13:28.611145 [**] [1:10000006:1] PORT SCAN - TCP FIN SCAN DETECTED [**] [Priority: 0] {TCP} 192.168.0.10:2776 -> 10.0.0.10:20
```

Task 4.5c: TCP Xmas Scan

```
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP XMAS SCAN DETECTED"; flags:FPU;
threshold: type both, track by_src, count 20, seconds 60; sid:10000007; rev:1;)
```

[28] Attack command:

```
$ sudo hping3 -UPF 10.0.0.10 -p ++1 -c 30
```

```

anujtr@client-vm:~$ sudo hping3 -UPF 10.0.0.10 -p ++1 -c 30
HPING 10.0.0.10 (enp0s1 10.0.0.10): FPU set, 40 headers + 0 data bytes
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=1 flags=RA seq=0 win=0 rtt=2.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=2 flags=RA seq=1 win=0 rtt=2.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=3 flags=RA seq=2 win=0 rtt=9.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=4 flags=RA seq=3 win=0 rtt=8.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=5 flags=RA seq=4 win=0 rtt=7.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=6 flags=RA seq=5 win=0 rtt=7.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=7 flags=RA seq=6 win=0 rtt=7.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=8 flags=RA seq=7 win=0 rtt=6.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=9 flags=RA seq=8 win=0 rtt=6.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=10 flags=RA seq=9 win=0 rtt=5.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=11 flags=RA seq=10 win=0 rtt=5.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=12 flags=RA seq=11 win=0 rtt=1.5 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=13 flags=RA seq=12 win=0 rtt=7.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=14 flags=RA seq=13 win=0 rtt=3.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=15 flags=RA seq=14 win=0 rtt=8.1 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=16 flags=RA seq=15 win=0 rtt=2.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=17 flags=RA seq=16 win=0 rtt=6.4 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=18 flags=RA seq=17 win=0 rtt=4.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=19 flags=RA seq=18 win=0 rtt=4.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=20 flags=RA seq=19 win=0 rtt=2.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=23 flags=RA seq=22 win=0 rtt=6.0 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=24 flags=RA seq=23 win=0 rtt=5.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=25 flags=RA seq=24 win=0 rtt=10.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=26 flags=RA seq=25 win=0 rtt=6.2 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=27 flags=RA seq=26 win=0 rtt=10.9 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=28 flags=RA seq=27 win=0 rtt=8.8 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=29 flags=RA seq=28 win=0 rtt=6.6 ms
len=40 ip=10.0.0.10 ttl=63 DF id=0 sport=30 flags=RA seq=29 win=0 rtt=6.1 ms

--- 10.0.0.10 hping statistic ---
30 packets transmitted, 28 packets received, 7% packet loss
round-trip min/avg/max = 1.5/6.1/10.9 ms

05/01-21:14:51.692409 [**] [1:10000007:1] PORT SCAN - TCP XMAS SCAN DETECTED" [**] [Priority: 0] {TCP} 192.168.0.10:2465 -> 10.0.0.10:20

```

Task 4.5d: TCP Null Scan

```

alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP NULL SCAN DETECTED"; flags:0;
threshold: type both, track by_src, count 20, seconds 60; sid:10000008; rev:1;)

```

[29] Attack command:

```
$ sudo nmap -sN 10.0.0.10 -p 1-100
```

```
anujtr@client-vm:~$ sudo nmap -sN 10.0.0.10 -p 1-100
Starting Nmap 7.80 ( https://nmap.org ) at 2026-05-01 21:15 UTC
Nmap scan report for 10.0.0.10
Host is up (0.0012s latency).
Not shown: 97 closed ports
PORT      STATE      SERVICE
21/tcp    open|filtered ftp
22/tcp    open|filtered ssh
80/tcp    open|filtered http

Nmap done: 1 IP address (1 host up) scanned in 1.35 seconds
```

```
05/01-21:15:41.915347  [**] [1:10000008:1] PORT SCAN - TCP NULL SCAN DETECTED [**] [Priority: 0] {TCP} 192.168.0.10:56548 -> 10.0.0.10:1
```

Task 4.5e: UDP Port Scan

```
alert udp any any -> $HOME_NET any (msg:"PORT SCAN - UDP SCAN DETECTED"; threshold: type
both, track by_src, count 20, seconds 60; sid:10000009; rev:1;)
```

[30] Attack command:

```
$ sudo hping3 -2 10.0.0.10 -p ++1 -c 30
```

```
anujtr@client-vm:~$ sudo hping3 -2 10.0.0.10 -p ++1 -c 30
HPING 10.0.0.10 (enp0s1 10.0.0.10): udp mode set, 28 headers + 0 data bytes
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1448 seq=0
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1449 seq=1
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1450 seq=2
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1451 seq=3
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1452 seq=4
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1453 seq=5
ICMP Port Unreachable from ip=10.0.0.10 name=UNKNOWN
status=0 port=1454 seq=6
```

```
05/01-21:16:52.905120  [**] [1:10000009:1] PORT SCAN - UDP SCAN DETECTED [**] [Priority: 0] {UDP} 192.168.0.10:2414 -> 10.0.0.10:20
```

V. CONCLUSION

This project provided hands-on experience with Network Intrusion Detection Systems (NIDS) using Snort, signature-based attack detection, and offensive security techniques. Key accomplishments include successfully installing and configuring Snort with custom detection rules, deploying nine different types of network attacks using hping3 and nmap, and verifying that all attacks were properly detected and logged by the IDS.

Important lessons learned include understanding the difference between signature-based and anomaly-based detection (Snort uses signatures/rules), the critical role of threshold detection in preventing alert flooding while still detecting attack patterns, and the importance of properly configuring network parameters (disabling reverse path filtering, enabling broadcast ping) to allow attack traffic for testing purposes.

A significant discovery was observing the amplification effect in the Smurf attack - a single ICMP packet to a broadcast address generated dozens of responses, demonstrating why broadcast amplification attacks are so effective. The project also highlighted how different TCP flags (ACK, FIN, NULL, Xmas) can be used for stealthy port scanning, each with different detection challenges.

Key technical insights include the necessity of disabling reverse path filtering (`rp_filter=0`) on the Gateway to allow spoofed packets through - without this, Land and Smurf attacks fail silently. Additionally, the importance of threshold-based detection became clear: without thresholds, a SYN flood would generate thousands of individual alerts, making the logs unusable. The threshold parameters (count 20, seconds 60) strike a balance between sensitivity and practicality.

One challenge encountered was that the Client VM initially lacked internet access for installing hping3. This was resolved by temporarily enabling NAT masquerading on the Gateway, demonstrating that project requirements sometimes conflict (attack isolation vs. tool installation). The NAT rules were left in place as they did not interfere with attack detection.

The main difference from Project 2 was the shift from implementing security controls (firewall rules) to implementing security monitoring (IDS rules). Project 2 focused on prevention through packet filtering and access control, while Project 3 focused on detection and logging of malicious activity. Both are essential layers of defense-in-depth security.

An important takeaway is that Snort's effectiveness depends entirely on the quality and coverage of its rules. The custom rules created for this project were specific to known attack patterns, meaning new or modified attacks could evade detection. This highlights the limitation of signature-based IDS - they can only detect known attack patterns and require constant rule updates to remain effective against evolving threats.

VI. APPENDIX B: ATTACHED FILES

Snort Local Rules File

Complete `/etc/snort/rules/local.rules`:

```
# Anuj T: Task 4.1 - Land Attack
alert tcp 10.0.0.10 80 -> 10.0.0.10 80 (msg:"LAND ATTACK DETECTED - Same Source and
Destination"; flags:S; sid:10000001; rev:1;)

# Anuj T: Task 4.2 - SYN Flood
alert tcp any any -> $HOME_NET any (msg:"SYN FLOOD ATTACK DETECTED"; flags:S; threshold:
type both, track by_dst, count 20, seconds 60; sid:10000002; rev:1;)

# Anuj T: Task 4.3 - Smurf Attack
alert icmp any any -> 192.168.0.255 any (msg:"SMURF ATTACK DETECTED - ICMP to
Broadcast"; itype:8; sid:10000003; rev:1;)

# Anuj T: Task 4.4 - UDP Flood
alert udp any any -> $HOME_NET any (msg:"UDP FLOOD ATTACK DETECTED"; threshold: type
both, track by_dst, count 10, seconds 60; sid:10000004; rev:1;)

# Anuj T: Task 4.5a - TCP ACK Scan
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP ACK SCAN DETECTED"; flags:A;
threshold: type both, track by_src, count 20, seconds 60; sid:10000005; rev:1;)

# Anuj T: Task 4.5b - TCP FIN Scan
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP FIN SCAN DETECTED"; flags:F;
threshold: type both, track by_src, count 20, seconds 60; sid:10000006; rev:1;)

# Anuj T: Task 4.5c - TCP Xmas Scan
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP XMAS SCAN DETECTED"; flags:FPU;
threshold: type both, track by_src, count 20, seconds 60; sid:10000007; rev:1;)

# Anuj T: Task 4.5d - TCP Null Scan
alert tcp any any -> $HOME_NET any (msg:"PORT SCAN - TCP NULL SCAN DETECTED"; flags:0;
threshold: type both, track by_src, count 20, seconds 60; sid:10000008; rev:1;)

# Anuj T: Task 4.5e - UDP Scan
alert udp any any -> $HOME_NET any (msg:"PORT SCAN - UDP SCAN DETECTED"; threshold: type
both, track by_src, count 20, seconds 60; sid:10000009; rev:1;)
```

System Configuration

Gateway `/etc/sysctl.conf` additions:

```
# Anuj T: Project 3 Settings
net.ipv4.ip_forward = 1
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.all.rp_filter = 0
net.ipv4.conf.all.accept_redirects = 1
net.ipv4.conf.all.send_redirects = 1
net.ipv4.icmp_echo_ignore_broadcasts = 0
```

VII. REFERENCES

- [1] Snort Users Manual 2.9.16, available at http://manual-snort-org.s3-website-us-east-1.amazonaws.com/snort_manual.html
- [2] Snort IDS Tools, available at <http://books.gigatux.nl/mirror/snortids/0596006616/main.html>
- [3] hping3 tutorial, available at <https://www.radarhack.com/tutorial/hping2.pdf>
- [4] hping documentation, available at <http://www.hping.org/documentation.php>

Video Link: <https://youtu.be/TGiQVUtNZEw>